

AI Utilization for Manufacturing Offices: Case Study of Excel Data Integration at Manufacturing Company

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ABSTRACT

This community engagement program was implemented at Manufacturing Company, a Japanese-affiliated steel components manufacturer, particularly in data management and administrative decision-making. The initiative aims to optimize Excel-based data integration and Artificial Intelligence (AI) adoption, especially through the use of ChatGPT as a productivity-enhancing tool for the finance division. Prior to the intervention, fragmented Excel files and non-standardized formats hindered forecasting accuracy, often resulting in overstocking or understocking of raw materials.

The program consisted of two key components: (1) integrating procurement and stock data using advanced Excel formulas and (2) enhancing employee capabilities through training in ChatGPT-assisted data operations. Pre-test and post-test assessments were conducted to measure changes in understanding, readiness, and perceived effectiveness, complemented by a self-assessment of program satisfaction. Results showed significant improvements in participants' understanding of AI concepts (+1.34), increased confidence in using ChatGPT, and greater efficiency in executing data-related tasks.

A perfect satisfaction score (4.93 out of 5) from the self-assessment indicated strong approval of the program's relevance, delivery, and applicability. This case study provides a practical blueprint for similar manufacturing companies in Indonesia seeking to bridge the digital transformation gap, particularly in administrative areas often overlooked during Industry 4.0 transitions. Moreover, the study highlights the growing need for upskilling in generative AI as a strategic asset in the modern industrial workforce.

KEYWORDS:

Artificial Intelligence (AI); Data Integration; ChatGPT; Manufacturing Operational; Digital Transformation

A. INTRODUCTION

The global manufacturing industry is undergoing a fundamental transformation driven by the adoption of Artificial Intelligence (AI). According to a study by Databricks and MIT (2024), 76% of industry players anticipate an efficiency increase of more than 25% within the next two years through the integration of this technology. This revolution affects not only production lines but also administrative and office functions, which serve as the backbone of strategic decision-

making. A report by Harvard Business Review (2024) reveals that 41% of manufacturing companies now allocate more than 25% of their technology budgets to data- and AI-driven solutions. However, 36% of them remain trapped in data silos due to legacy systems and the lack of data format standardization, which hinders the potential to generate business value. One of the most pressing issues is the absence of an integrated information system, particularly in managing data for raw material forecasting (Van Belle et al., 2021).

One of the latest innovations accelerating digital transformation is Generative AI, such as ChatGPT, which is increasingly viewed as a disruptor in managerial functions. As a large language model, ChatGPT is capable of providing analysis, recommendations, and data-driven assistance that can enhance the effectiveness of decision-making processes (Ikeda, 2024). In the context of manufacturing administration, spreadsheet processing, particularly with Excel, has emerged as a major pain point. A significant portion of analysts' time is consumed by data preprocessing tasks such as format normalization and file consolidation rather than by strategic analysis (Pradhan et al., 2025). This situation creates a paradoxical productivity gap, even though modern Generative AI solutions have demonstrated their ability to reduce administrative data processing time through the automation of complex dataset integration (Alavi et al., 2024).

A steel components manufacturer and part of a Japanese manufacturing group, faces similar challenges. These include the presence of dozens of Excel files in inconsistent formats and lacking integration, which hinder the speed and accuracy of material purchasing forecasts. This challenge is further complicated by the company's Japanese corporate culture, which tends to be skeptical about using public AI platforms such as ChatGPT, primarily due to concerns related to data security (Ikkatai et al., 2024). Nonetheless, research has shown that Generative AI has significant potential to support administrative tasks within manufacturing environments (Fatima and Rahima, 2024). This technology is capable of automating routine processes, strengthening knowledge management, and accelerating planning and production control processes (Ghobakhloo et al., 2024). In this context, the collaboration with Manufacturing Company is designed to address real-world operational challenges through the development of an efficient Excel data integration system combined with AI-supported tools for data-driven decision-making.

Strategically, this initiative is expected to serve as a replicable blueprint for other manufacturing industries in Indonesia to help bridge the digital divide. This is especially relevant during the transition toward comprehensive industrial data utilization, which is critical in the era of Industry 4.0. However, according to the World Economic Forum (2022), only 39% of manufacturing companies have been able to scale their data initiatives beyond a single product line, indicating that the full potential of industrial data remains underutilized. This vision also aligns with projections from BCG (2025), which emphasize the need to reskill 50% of the workforce in Generative AI technologies within the next three years as a foundation for AI-first business transformation. In response to these challenges, this community engagement program is designed with two main objectives: (1) to increase AI literacy among staff through a program focused on optimizing the use of ChatGPT, and (2) to develop an adaptive Excel data integration system that supports more efficient decision-making within Manufacturing Company.

B. METHOD

This service activity was carried out at Manufacturing Company for four months, from March to June 2025. The activity is divided into two main components: (1) data integration using Excel, and (2) optimizing the use of AI through the use of ChatGPT for administrative staff. Prior to the activities, the company had started using ChatGPT and Deepseek, but it was still limited to managers. In addition, operational data was still scattered in various files, not interconnected, and not updated in real-time, which led to overstock and stock shortages.

2.1 Participants and Data Collection Techniques

A total of 17 staff from the Finance Division participated in this activity. They were selected using purposive sampling technique based on their involvement in procurement, purchase order (PO), warehouse stock, and shipping management activities. Data collection was conducted through: (1) Direct observation during the program. (2) Pre-test and post-test through Google Form. (3) Self-assessment after the program is implemented. (4) Internal company documents (Excel files & participants' work results).

2.2 Data Integration

Microsoft Excel was used as the main tool to manage, integrate, and analyze the company's operational data. The data analyzed includes procurement, ETA, forecast, stock, purchase order (PO), and delivery. Excel facilitates the systematic organization of data through the use of formulas such as SUMIF, COUNTIF, IF, and VLOOKUP. Dynamic table features and data visualization also support data-driven decision making. The implementation of this activity includes (1) Data Input with the collection of the main file that includes the analyzed data. (2) Data Cleaning which includes adjusting date, number, and text formats; removing duplicates; handling blank values; and normalizing the writing of material codes to be uniform across files. (3) Data Integration from various sources into one main base using formulas such as VLOOKUP, INDEX-MATCH, and IFERROR. (4) Analysis and Automation of data that has been integrated with formulas such as SUMIF, COUNTIF, and IF. Analysis includes sales forecast, PO status, delivery progress, safety stock calculation, order requirement estimation, and lead time. ChatGPT is used as an AI assistant to help construct complex formulas and support data-driven decision making.

2.3 AI Optimization through ChatGPT

AI optimization in the manufacturing sector drives automation, increased productivity, and more accurate decision-making. One application is data integration in administration, using Microsoft Excel combined with ChatGPT as an AI assistant. This approach is complemented by four methods: (1) a pre-test to measure participants' initial understanding of the basic concepts of AI and ChatGPT, their usage experience, and readiness to learn new technologies. The results of the pre-test were used to adjust the program strategy, (2) implementation or providing the results of data integration to Manufacturing Company for study and review, (3) post-test to evaluate the

improvement of participants' understanding in composing prompts, understanding how ChatGPT works, and applying AI in daily work, and (4) self-assessment to assess participants' satisfaction with the program. The assessment included the relevance of the material, the effectiveness of the learning method, and its impact on work efficiency, data input accuracy, and prompt-based logical analysis skills.

2.4 Data Analysis Technique

Data analysis techniques were carried out descriptively quantitatively and qualitatively. Pre-test and post-test data were analyzed by comparing the average scores to see the improvement of participants' understanding. Meanwhile, self-assessment data was analyzed thematically to evaluate participants' perceptions of the effectiveness of the program and its impact on work efficiency.

C. RESULT AND DISCUSSION

3.1 Data Integration

Before the program was conducted, operational data management at Manufacturing Company was in a separate condition. Each process such as procurement, ETA, sales forecast, warehouse stock, and purchase order (PO) was managed separately using Excel files that did not have a standard format. This lack of integration causes low visibility of information, slows down the work process, and leads to inaccurate decisions. Overstocks led to freezing of funds, while shortages posed a risk of delayed deliveries and potential loss of customers. In addition, forecast data from customers was not fully integrated with the Master Production Schedule and production system, so information was often out of sync between departments.

Initial discussions with management and Finance staff became the main foundation in designing a system that was relevant and applicable to the needs of the field. Documentation of these activities is shown in Figure 1.



Figure 1. Documentation of initial discussion with management

As a response to the problem, all scattered files were consolidated into one main Microsoft Excel-based workbook. The workbook was designed with tabs per product category, and features a validation system, dynamic formula structure, and monitoring dashboard. Formulas such as VLOOKUP, INDEX-MATCH, IFERROR, AND, and IF were used to automate calculations such as stock *balance*, availability status, and recommended actions.

The screenshot shows an Excel spreadsheet with a table titled 'Rekomendasi Pengadaan Material (Periode Jan-Mar 2026)'. The table has columns: Item Code, Date, Balance, Stock Ratio, Status, Minus, and Rekomendasi. Red boxes highlight various formulas and data points:

- Formula in cell E3: `=IF(C3<0,"Leaking";IF(C3<20,"Warning";"OK"))`
- Formula in cell E4: `=IF(C3<0,"Minus stok "&C3;IF(C3<20,"Stok hampir habis";"Stok aman"))`
- Formula in cell E5: `=IF(AND(C3<0,D3<0.5),"Segera tambah stok"; IF(AND(C3<0,D3<1),"Pertimbangkan PO tambahan"; IF(C3<=-100,"Prioritaskan urgent delivery"))`
- Formula in cell E6: `=IF(AND(C3<0,D3<0.5),"Segera tambah stok"; IF(AND(C3<0,D3<1),"Pertimbangkan PO tambahan"; IF(C3<=-100,"Prioritaskan urgent delivery"))`
- Formula in cell E7: `=IF(AND(C3<0,D3<0.5),"Segera tambah stok"; IF(AND(C3<0,D3<1),"Pertimbangkan PO tambahan"; IF(C3<=-100,"Prioritaskan urgent delivery"))`
- Formula in cell E8: `=IF(AND(C3<0,D3<0.5),"Segera tambah stok"; IF(AND(C3<0,D3<1),"Pertimbangkan PO tambahan"; IF(C3<=-100,"Prioritaskan urgent delivery"))`
- Formula in cell E9: `=IF(AND(C3<0,D3<0.5),"Segera tambah stok"; IF(AND(C3<0,D3<1),"Pertimbangkan PO tambahan"; IF(C3<=-100,"Prioritaskan urgent delivery"))`
- Formula in cell E10: `=IF(AND(C3<0,D3<0.5),"Segera tambah stok"; IF(AND(C3<0,D3<1),"Pertimbangkan PO tambahan"; IF(C3<=-100,"Prioritaskan urgent delivery"))`
- Formula in cell E11: `=IF(AND(C3<0,D3<0.5),"Segera tambah stok"; IF(AND(C3<0,D3<1),"Pertimbangkan PO tambahan"; IF(C3<=-100,"Prioritaskan urgent delivery"))`
- Formula in cell E12: `=IF(AND(C3<0,D3<0.5),"Segera tambah stok"; IF(AND(C3<0,D3<1),"Pertimbangkan PO tambahan"; IF(C3<=-100,"Prioritaskan urgent delivery"))`
- Formula in cell E13: `=IF(AND(C3<0,D3<0.5),"Segera tambah stok"; IF(AND(C3<0,D3<1),"Pertimbangkan PO tambahan"; IF(C3<=-100,"Prioritaskan urgent delivery"))`
- Formula in cell E14: `=IF(AND(C3<0,D3<0.5),"Segera tambah stok"; IF(AND(C3<0,D3<1),"Pertimbangkan PO tambahan"; IF(C3<=-100,"Prioritaskan urgent delivery"))`
- Formula in cell E15: `=IF(AND(C3<0,D3<0.5),"Segera tambah stok"; IF(AND(C3<0,D3<1),"Pertimbangkan PO tambahan"; IF(C3<=-100,"Prioritaskan urgent delivery"))`

Figure 2. Excel file that has been integrated

This system transformation has resulted in real efficiency in the daily work process. The Finance team can now monitor stock in real-time, calculate order requirements based on actual data, and determine purchasing priorities through just one integrated page. Processes that previously took hours can now be completed in minutes. ChatGPT was also utilized as a digital assistant to generate complex formulas and provide contextual recommendations based on specific operational conditions. The automatic recommendation feature displayed in the workbook allows staff to make quick and measurable decisions, especially in critical stock situations that require immediate action. The perceived effectiveness of this data integration system also confirms the argument of Ghobakhloo et al. (2024) that the integration of Generative AI into manufacturing administrative processes can accelerate digital adoption in non-productive lines, which have tended to be neglected in the industrial transformation 4.0 agenda.

3.2 AI Optimization through ChatGPT

This program was conducted to improve the understanding and skills of manufacturing industry employees in conducting data integration by optimizing the use of Artificial Intelligence (AI) as an assistant in data-based decision making. To evaluate the effectiveness of the results of this program, a pre-test was conducted to measure the level of understanding and expectations of employees, which will be compared with the results of the post-test after the program is completed. The assessment was conducted to one related division as a representative using several indicators relevant to the program which can be seen in the following table.

Table 1. Pre-test and Post-test Results

Indicator	Average pre-test Score	Average post-test Score	Difference
Basic understanding of AI	3,33	4,67	+1,34
Knowledge of ChatGPT	4,00	4,67	+0,67
Experience integrating data with AI	4,33	5,00	+0,67
Expectations/Evaluation of the program	4,00	5,00	+1,00

Table 1 presents the results of the pre-test and post-test assessment of the four main indicators, each of which has 3 questions. The scores in the table are the average of the scores collected from the relevant divisions. Based on the above results, it was found that there was an increase in the average score on 4 indicators after the implementation of this program. The most significant increase occurred in the Basic Understanding of AI indicator by 1.34 points. This indicates that employees have a better understanding of basic AI after participating in this program. Overall, the relatively consistent increase in all indicators is a measure of the success of the program provided to the company in having a positive impact on related divisions.

Of the total 5 indicators of the pre-test and post-test assessment, 4 of them were used and measured in both types of tests. However, each test has 1 relevant indicator used specifically in each test only. The results of these indicators are shown in Table 2. Pre-test Specific Indicators and Table 3. Post-test Specific Indicators below.

Table 2. Pre-test Specific Indicators

Indicator	Average Post-test Score
Readiness to learn new technology	4,33

Table 2 shows that the pre-test uses a special indicator, namely Readiness to Learn New Technology, which has an average value of 4.33 on a scale of 1-5. The high value obtained from this indicator shows the high enthusiasm and readiness of the relevant division employees to develop their competence in using the data integration system with AI assistance before the program implementation.

Table 3. Post-test Specific Indicators

Indicator	Average Post-test Score
Effectiveness of AI-based data integration	4,33

In table 3 above, it is known that the post-test uses a special indicator, namely the Effectiveness of AI-Based Data Integration, which gets a perfect average score of 5.00 on a scale of 1-5. The results above show that the implemented program is very effective in helping related divisions understand and implement AI-based data integration. In addition, this high effectiveness also shows that this program is able to have a very positive impact on employees in supporting work efficiency and can facilitate data-based decision making.

The difference in indicators in the pre-test and post-test is the impact of adjusting the context of the assessment instrument which is readjusted to the needs of each stage. Although the two indicators cannot be compared directly with quantitative methods, they still provide an important picture of employee behavior and perceptions of the program. The presentation of data in comparative and descriptive form aims to provide a deeper understanding of the impact of the program, both in terms of the initial readiness of participants and the results felt by employees after participating in this program. Therefore, despite having different indicators, the two indicators complement each other and evaluate the success of this program.

This finding reinforces the results of Ikeda's study (2024) which states that hands-on experience and self-exploration are the most effective approaches in improving the literacy of using Generative AI in the work environment. The increased understanding and confidence of staff in using ChatGPT suggests that initial resistance to AI can be overcome through practical and contextual approaches.

A successful program is defined not only by the achievement of its intended objectives but also by the satisfaction level of its partner stakeholders. To determine the level of partner satisfaction with the program we have implemented, we sent a self-assessment questionnaire that included several aspects of assessment, including: (1) Relevance of the program to work needs. (2) Quality of team mentoring. (3) Effectiveness of AI-based data integration technology solutions. (4) General satisfaction with the overall program. (5) Intention to recommend the program to others.

Table 4. *Self-Assessment Results*

Assessment Variable	Average Score
Relevance of the program to work needs	4,67
Quality of team mentoring	5,00
Effectiveness of AI-based data integration technology solutions	5,00
General satisfaction with the overall program	5,00
Intention to recommend the program to others	5,00
Overall Average	4.93

Based on the self-assessment results, the overall average score of the partners' satisfaction level is 4.93 on a scale of 1-5. The 4.93 score on this self-assessment not only represents the technical success of the program, but also supports the findings of Pradhan et al. (2025) who emphasized the importance of pre-analysis and system integration in optimizing the potential of administrative data for decision-making processes. The fact that participants rated the effectiveness of AI at 5.00 indicates the industry's readiness to shift from manual-based processes to logic-based automation.

In general, the quality of team mentoring, effectiveness of AI-based data integration technology solutions, general satisfaction with the overall program, and intention to recommend the program to others received the highest ratings in this self-assessment. Meanwhile, the assessment of the program's relevance to work needs received relatively lower scores than other aspects of the assessment. Thus, it can be concluded that partners are very satisfied with the implementation of this program with more attention to the relevance of the program to be delivered to the work needs of partners.

In addition to the assessment, partners also provided notes in the form of constructive criticism and suggestions for evaluation and further program development through this self-assessment. Overall, the feedback provided by the partners emphasized the importance of preparing structured program documents as a basis for director decision-making at the managerial level, as well as support for us to continue to conduct research and innovation because the contribution of the younger generation's thinking is needed in today's industry.

As such, this study not only concretely demonstrates how AI and data integration can be applied in the field, but also contributes to addressing the gap highlighted by the World Economic Forum (2022) regarding the limited scalability of data initiatives in the manufacturing sector. This study shows that a simple yet targeted approach can effectively drive the digitization of administrative lines and have a tangible impact on operational efficiency.

D. CONCLUSION

This community service activity has successfully addressed the real challenges faced by Manufacturing Company in managing operational data administration, which is fragmented and poorly integrated, which is fragmented and less integrated, especially in the context of material purchase planning. Through two main approaches, namely Microsoft Excel-based operational data integration and Artificial Intelligence (AI) optimization program using ChatGPT, this program is able to improve the work efficiency of finance division staff in managing procurement, forecast, stock, and purchase order information in a more structured, accurate, and real-time manner.

The pre-test and post-test results showed a significant increase in participants' understanding of AI concepts and their application in work, with the highest increase in the indicator "basic understanding of AI" (+1.34 points). In addition, the very high self-assessment scores (average 4.93 out of 5) indicated that the partners were very satisfied with the relevance of the materials, the quality of the mentoring, the effectiveness of the technology solutions, and the positive impact on daily work efficiency.

Overall, this activity proves that the utilization of generative AI such as ChatGPT can make a significant contribution in supporting the digital transformation of the manufacturing sector, especially in managing operational data that has often been overlooked. This case study also reinforces the importance of increasing AI literacy and strengthening digital capacity among non-technical staff as a strategic step towards industry readiness in facing the AI-First era.

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CONFLICT OF INTEREST

In the preparation of this article, there are no major conflicts of interest with the company. Manufacturing Company provided dummy data as sample data and also general insights into the company's needs and expectations of the dashboard design in the data integration program that we will implement. In this article, the company is only the subject of the study through the completion of pre-test, post-test, and self-assessment of partner satisfaction levels to support the evaluation of the program that has been implemented. The company was not involved in the study design, data analysis, or decision making in the making or publishing of this article.

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